RenAI – MVP Product Requirements Document

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1. PRODUCT OVERVIEW

RenAI is a B2B web-based sustainability operations platform designed to help industrial companies centralize environmental data that is typically fragmented across multiple departments. The platform will allow companies to manage waste flows, carbon footprint data, documentation, regulations, team collaboration, and sustainability planning in one place. The MVP will be validated first with Maxtil, a textile manufacturing company.

1.1. Product Purpose

RenAI aims to give companies a clear and centralized view of their sustainability-related data, including waste generation, greenhouse gas emissions, supporting documentation, and operational actions. The platform should reduce time spent gathering information, improve internal coordination, support audit preparation, and help companies monitor reduction opportunities.

1.2. Product Type

B2B multi-tenant web application for industrial and manufacturing companies.

1.3. Initial Customer Profile

The first customer for MVP validation is Maxtil, a textile manufacturing company. The platform is being designed primarily for manufacturing and industrial business that need to organize environmental data across ESG, waste, energy, operations and compliance-related functions.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Industrial companies usually have sustainability-related data spread across different departments, systems, spreadsheets, emails, and documents. Information related to waste, fuel consumption, electricity use, emissions, and compliance is often managed separately by different teams, which makes it difficult to have a complete and reliable overview of the company's environmental performance.

This fragmentation creates several problems:

- Difficulty accessing all relevant information in one place,
- Time lost collecting data manually,
- Lack of visibility across departments,
- Weak traceability of supporting documentation,
- Difficulty preparing for audits,
- Inconsistency in emissions calculations,
- Reduced ability to plan improvements and reduction actions.

RenAI is being created to solve this problem by centralizing sustainability data, documentation, emissions tracking, and operational follow-up in a single platform.

2.2. Core Problem Being Solved

The core problem RenAI solves is the lack of a centralized system where companies can organize, monitor, and analyze environmental and sustainability data across the business.

2.3. Product Objective

The objective of the MVP is to provide a web-based platform that allows a company to:

- Centralize environmental records and documentation,
- Manage waste flow information,
- Record carbon footprint activity data,
- Calculate emissions across the relevant GHG Protocol scopes,
- Monitor key sustainability indicators,

- Support audit preparation,
- Improve internal collaboration between teams.

2.4. Business Objective

The business objective of the MVP is to validate that companies are willing to use RenAI as their operational sustainability platform and that the product creates enough practical value to support future adoption by other industrial companies in Portugal and later in Europe.

2.5. User Value Proposition

The platform should deliver three main benefits to users:

- Save time,
- Provide a clear overview of environmental data,
- Keep all relevant information in one place.

2.6. Long-Term Vision

In the long term, RenAI is intended to become a trusted sustainability operations platform that companies rely on for environmental data management, emissions tracking, documentation, analysis, planning, and compliance support.

3. TARGET USERS AND CUSTOMER PROFILE

3.1. Target Customer Type

RenAI is a B2B platform designed for industrial and manufacturing companies that need to manage sustainability-related data in a centralized and structured way. The initial focus is on companies that generate and manage significant operational environmental data, especially in areas such as:

- Waste generation,
- Fuel consumption,
- Electricity consumption,
- Greenhouse gas emissions,
- Supporting compliance documentation,
- Sustainability reporting preparation.

3.2. Initial Validation Customer

The first company expected to validate the MVP is **Maxtil**, a textile manufacturing company.

This means the first version of the platform should be especially suited to the operational reality of a manufacturing business, including multiple internal data sources, recurring documentation, and the need to monitor both waste and carbon footprint information.

3.3. Primary User Groups

The main user groups of the platform are expected to include:

- ESG or Sustainability teams,
- Waste management teams,
- Fuel and energy teams,
- Operations teams,
- Plant or site managers,
- Internal collaborators responsible for data entry and documentation,
- Management users who need visibility over sustainability performance.

3.4. User Needs

These users need a system that allows them to:

- Access sustainability-related information in one place,
- Register and update operational data,
- Upload and retrieve supporting files,
- Monitor emissions and waste-related KPIs,
- Analyze sustainability performance,
- Collaborate internally with other team members,
- Prepare for audits and reporting processes more efficiently.

3.5. Future Customer Expansion

After validating the MVP with the first customer, RenAI should be positioned to expand to other industrial businesses in Portugal and later in other European markets.

The platform should therefore be designed from the beginning with a multi-tenant structure and with enough flexibility to support companies from different industrial sectors, even if the first implementation is tailored to a textile manufacturer.

3.6. Customer Environment Assumption

The product assumes that customer companies may already manage part of their sustainability data through spreadsheets, PDFs, invoices, internal reports, and fragmented operational processes. RenAI should replace this fragmented approach with a single structured platform.

4. PRODUCT SCOPE AND MVP GOALS

4.1. MVP Goal

The goal of the MVP is to deliver a web-based platform that allows industrial companies to centralize and manage their main sustainability data in one place, with an initial focus on waste management, carbon footprint tracking, documentation, team collaboration, and audit preparation.

The MVP should be functional, visually strong, user-friendly, and structured in a way that supports future growth.

4.2. Primary Goals of the MVP

The MVP should allow users to:

- Centralize sustainability-related information,
- Register and manage waste flows,
- Record emissions-related activity data,
- Calculate greenhouse gas emissions,
- Upload and organize supporting documentation,
- Monitor key performance indicators,
- Analyze environmental performance through graphs and visualizations,
- Assign and track internal tasks,
- Prepare more efficiently for audits and compliance-related reviews.

4.3. Business Goals of the MVP

From a business perspective, the MVP should:

- Validate the product with a real industrial customer,
- Prove that companies find value in a centralized sustainability platform,
- Create a strong product foundation for future commercial growth,
- Be scalable enough to support additional customers and modules later,
- Present a polished and credible solution.

4.4. Included in MVP

The MVP should include the following core modules and capabilities:

- Dashboard
- Waste Flows
- Carbon Footprint
- Analysis
- Documentation
- Regulations
- Team Overview

It should also include:

- User authentication,
- Multi-tenant company structure,
- Role-based access,
- Document uploads and downloads,
- Activity history,
- Task assignment by Admin users,
- KPI monitoring,
- Emissions calculations,
- Audit-friendly traceability.

4.5. Excluded or Limited in MVP

To keep the MVP realistic and focused, the following items should not be treated as full core features in the first version:

- Advanced predictive AI,
- Fully automated compliance engine,
- Complex ERP integrations,
- Real-time IoT integrations,
- Native mobile app,
- Highly advanced workflow automation,
- Full legal automation for all EU countries.

These can be added in later phases once the product is validated.

4.6. Product Priorities

The main priorities for the MVP are:

- Strong and beautiful user experience,
- Clear and intuitive platform structure,
- Reliable core functionality,
- Scalability,
- Practical usefulness for the customer from day one.

4.7. Success Criteria for the MVP

The MVP can be considered successful if it allows the first customer to:

- Use the platform as a central place for sustainability data,
- Manage waste and carbon-related records efficiently,
- Access supporting documentation easily,
- Gain a clearer overview of environmental performance,
- Improve internal coordination,
- See enough value to continue using the platform and support its evolution.

5. USER ROLES AND PERMISSIONS

5.1. Purpose

The platform should include a role-based access structure to ensure that different users can only access and perform the actions appropriate to their responsibilities.

For the MVP, three user roles should be implemented:

- Admin
- Collaborator
- Viewer

5.2. Audit and Accountability

For accountability and traceability, the system should record which user created or updated important records and documents. This is especially important for Admin and Collaborator actions.

6. PLATFORM STRUCTURE AND NAVIGATION

6.1. Purpose

The platform should have a clear, simple, and modern navigation structure that allows users to move easily between the main functional areas of the system. The navigation should reflect the main operational needs of the customer and keep the product intuitive from the first use.

6.2. Main Navigation Modules

The MVP should include the following main sections in the sidebar navigation:

- Dashboard
- Waste Flows
- Carbon Footprint
- Analysis
- Documentation
- Regulations
- Team Overview

6.3. Navigation Principles

The platform navigation should follow these principles:

- Simple and easy to understand,
- Minimal and clean,
- Consistent across all pages,
- Designed for fast access to the most important modules,
- Structured in a way that can scale with future modules.

The product should feel like a modern SaaS application with a minimalist and premium interface.

6.4. Module Intent

Each navigation section should have a clear purpose:

Dashboard
High-level overview of the company's sustainability activity, key KPIs, latest entries, and recent team actions.

Waste Flows
Operational area for creating, managing, and reviewing all waste flow records.

Carbon Footprint
Main data-entry and management area for Scope 1, Scope 2, Scope 3, and Production Emission Factor records.

Analysis
Advanced visual analytics area with configurable graphs, filters, and deeper exploration of carbon and sustainability data.

Documentation
Central repository for all files uploaded across the platform, with search, filter, preview, and download capabilities.

Regulations
Area for relevant EU and country-level sustainability and environmental regulations, including curated updates and AI-assisted news support.

Team Overview
Area for user management visibility, team activity tracking, and task monitoring.

6.5. Carbon Footprint Internal Structure

Within the Carbon Footprint module, there should be internal navigation or tabs for the following subsections:

- Scope 1
- Scope 2
- Scope 3
- Production Emission Factor

This structure keeps the main sidebar clean while organizing the carbon footprint area clearly.

6.6. Suggested Global Page Elements

Across the platform, the following global elements should be present where appropriate:

- Sidebar navigation,
- Top header,
- Page title,
- Search and filter area,
- Primary action button,
- Notifications or alerts area,
- User profile/account menu.

6.7. Search and Filter Logic

Modules with high data volume should support internal search and filtering. This is especially important in:

- Waste Flows,
- Carbon Footprint,
- Documentation,
- Analysis,
- Team Overview.

6.8. Future Scalability

The navigation structure should allow future addition of modules without forcing a redesign of the whole platform. For example, future sections may include:

- Settings,
- Reports,
- Integrations,
- AI Assistant,
- Notifications.

These do not need to be core sidebar items in the MVP, but architecture should allow them later.

6.9. UX Recommendation

The sidebar should remain visually light and uncluttered. The main Dashboard should stay focused on overview and recent activity, while detailed visual exploration should happen in the Analysis section, not in the main Dashboard.

7. DASHBOARD MODULE

7.1. Purpose

The Dashboard should provide a clean and immediate overview of the company's sustainability activity at a high level.

Its purpose is not to provide deep analysis, but rather to give users a quick summary of the most important indicators, recent updates, and team activity.

7.2. Dashboard Role in the Product

The Dashboard should function as the platform's main overview page.

It should help users quickly understand:

- The current sustainability status of the company,
- The most important KPI summaries,
- Whether there are missing items or pending actions,
- What has recently changed in the platform,

- What the team has been doing.

Detailed graph exploration and advanced emissions analysis should be handled in the **Analysis** module, not in the Dashboard.

7.3. Dashboard Design Principles

The Dashboard should be:

- Clean,
- Minimal,
- Visually strong,
- Easy to read,
- Not overloaded,
- Focused on summary information.

It should feel modern and premium, with good spacing, clear card hierarchy, and a strong sense of organization.

7.4. Main Dashboard Components

The Dashboard should include the following types of sections:

- KPI summary cards,
- Recent activity section,
- Latest team actions,
- Pending or open tasks,
- Data completeness or alert section,
- Quick operational overview.

7.5. Recommended KPI Summary Cards

The top of the Dashboard should display the most relevant high-level KPI cards. Recommended cards include:

- Total Waste Flows
- Estimated Waste Volume
- Hazardous Waste Flows
- Total Scope 1 Emissions
- Total Scope 2 Emissions

- Total Scope 3 Emissions
- Production Emission Factor

These should be displayed in a simple and visually balanced card layout.

7.6. Recent Activity Section

The Dashboard should include a section showing recent platform activity, such as:

- Newly created waste flows,
- Newly added carbon footprint entries,
- Recently uploaded documents,
- Recently updated records,
- Recently completed tasks.

This helps users understand what has changed since their last visit.

7.7. Latest Team Actions

A dedicated area should show the latest actions performed by team members, such as:

- Who added a new record,
- Who updated an existing entry,
- Who uploaded documentation,
- Who completed a task,
- Who was assigned a new task.

This strengthens visibility and internal coordination.

7.8. Open Task Overview

The Dashboard should include a summary of open tasks, especially for Admin users.

This section should show:

- Number of open tasks,
- Overdue tasks,
- Tasks due soon,

- Tasks by status,
- Tasks by assigned user.

This should remain a summary view, with full detail accessible in Team Overview.

7.9. Data Quality / Completeness Summary

The Dashboard should include a light overview of data quality or completeness, for example:

- Entries missing documentation,
- Incomplete emissions records,
- Waste flows missing required classifications,
- Pending items requiring review.

This section should help users identify operational gaps without turning the Dashboard into a detailed compliance page.

7.10. Time and Filter Logic

The Dashboard should support at least basic filtering by:

- Reporting period,
- Plant or location, if applicable,
- Company-level overview within the tenant.

This allows users to see the overview for a selected time period instead of only a fixed all-time view.

7.11. What Should Not Be in the Dashboard

The Dashboard should not become the main page for detailed graph exploration.

The following should be limited or avoided here and belong in the Analysis module.

- Too many complex charts,
- Deep carbon analysis,
- Configurable graph libraries,
- Overloaded comparison views,
- Large analytical tables.

7.12. Empty State Logic

If the company has no data yet, the Dashboard should present helpful empty states that guide the user toward the first actions, such as:

- Create a waste flow,
- Add a Scope 1 or Scope 2 entry,
- Upload documentation,
- Invite collaborators.

This is especially important for onboarding the first users.

7.13. Product Intent

The Dashboard should feel like the company's sustainability control center at a glance: clear, calm, actionable, and operationally useful.

8. WASTE FLOWS MODULE

8.1. Purpose

The Waste Flows module should be one of the main operational areas of the platform.

Its purpose is to allow the company to register, organize, and monitor all waste streams generated in its operations, while keeping the related data and supporting documentation centralized and easy to access.

8.2. Module Objective

This module should allow users to:

- create waste flow records,
- maintain an organized overview of all waste streams,
- classify waste correctly,
- monitor generation frequency and estimated quantities,
- attach supporting documentation,
- prepare information for operational follow-up and audits.

8.3. Waste Flow Record Structure

Each waste flow record should contain the following fields.

8.3.1 General Information

- Waste Flow Name
- Category
- Status
- Waste Code (LoW / EWC)
- Description
- Material Composition

8.3.2 Quantities and Generation

- Estimated Quantity
- Unit
- Frequency of Generation

8.3.3 Operational Information

- Current Storage Method
- Current Destination
- Current Operator
- Plant / Location
- Internal Code
- Treatment Type
- Treatment Notes

8.3.4 Additional Information

- Recovery Potential Notes
- Notes
- Hazardous Material flag
- Priority flag

8.4. Waste Flow Entry Logic

When creating a waste flow, the user should be able to input all the relevant operational information in a structured form.

The form should be easy to complete and clearly separated into logical sections, so that the user does not feel overwhelmed.

The system should support both required and optional fields, with the most critical data being mandatory for record creation.

8.5. Recommended Required Fields

For the MVP, the following fields should be required when creating a waste flow:

- Waste Flow Name
- Category
- Waste Code (LoW / EWC)
- Estimated Quantity
- Unit
- Frequency of Generation
- Plant / Location, if the company uses more than one site
- Status

Other fields can remain optional but strongly recommended.

8.6. Waste Flow Actions

Users with the appropriate permissions should be able to:

- Create a new waste flow,
- Edit an existing waste flow,
- Archive a waste flow,
- View waste flow details,
- Attach documents to a waste flow,
- Download linked documents,
- Search for waste flows,
- Filter waste flows,
- Mark a waste flow as hazardous,
- Mark a waste flow as priority.

8.7. Waste Flows Overview Page

The main Waste Flows page should show a structured overview of all waste flow records.

It should include:

- Search bar,
- Filters,
- Create new waste flow button,
- Result count,
- List or table view of waste flows.

8.8. Recommended Waste Flow Filters

The page should support filtering by:

- Category,
- Status,
- Hazardous flag,
- Priority flag,
- Plant/location,
- Waste code,
- Treatment type,
- Frequency.

8.9. Recommended Waste Flow List Columns

The list view should display the most relevant columns, such as:

- Waste Flow Name
- Category
- Waste Code
- Plant / Location
- Estimated Quantity
- Unit
- Frequency
- Treatment Type
- Status

- Hazardous
- Priority
- Last Updated

8.10. Waste Flow Detail Page

Each waste flow should have its own detail view where the user can see all information associated with that waste stream.

This detail page should display:

- Full waste flow information,
- Linked documents,
- Activity history,
- Related tasks,
- Creation date,
- Last updated date,
- Created by,
- Updated by.

8.11. Documentation Linkage

A waste flow record should allow users to upload and attach relevant supporting files directly from within the record.

These documents should also appear in the central Documentation module.

Examples of linked documents may include:

- Waste collection documents,
- Operator records,
- Treatment evidence,
- Invoices,
- Transport documentation,
- Internal files,
- Audit evidence.

8.12. Hazardous Waste Logic

The module should support a hazardous material indicator, allowing waste streams to be clearly identified as hazardous where applicable.

This is important both for visibility and for future reporting and compliance workflows.

8.13. Treatment Classification Logic

The waste flow record should include a treatment type field that allows the user to identify how the waste is managed.

This should support treatment classification in line with recovery and disposal logic, so that users can distinguish between different waste handling pathways.

8.14. Operational Usefulness

The Waste Flows module should not function only as a static database. It should give the user a useful operational overview of the company's waste streams and make it easier to identify priorities, improvement opportunities, and missing information.

8.15. UX Recommendation

This module should feel structured, clean, and highly usable. Because waste flow records may contain many fields, the interface should rely on:

- Grouped form sections,
- Clean spacing,
- Simple labels,
- Intuitive filters,
- Clear detail pages.

8.16. Product Intent

The Waste Flows module should become the company's central operational registry for waste generation and waste management information.

9. CARBON FOOTPRINT MODULE OVERVIEW

9.1. Purpose

The Carbon Footprint module should be the central area of the platform for recording, calculating, and managing greenhouse gas emissions data.

Its purpose is to allow the company to organize emissions-related activity data in a structured way, calculate emissions using appropriate factors, and monitor carbon performance over time.

9.2. Module Objective

This module should allow users to:

- Record emissions-related activity data,
- Calculate emissions across the relevant GHG Protocol scopes,
- Organize supporting documentation,
- Monitor emissions by source and period,
- Compare results across locations and categories,
- Support audit preparation and internal analysis,
- Connect emissions data with production output through an emission intensity metric.

9.3. Internal Module Structure

The Carbon Footprint module should be divided into four internal sections:

- Scope 1
- Scope 2
- Scope 3
- Production Emission Factor

This structure keeps the module clear and aligned with the standard emissions accounting logic used by companies.

9.4. Role of Each Section

Scope 1
For direct emissions from sources owned or controlled by the company.

Scope 2
For indirect emissions from purchased electricity, steam, heat, or cooling, including both location-based and market-based calculation views.

Scope 3
For relevant indirect value-chain emissions categories, especially those most relevant to the customer's industry.

Production Emission Factor
For calculating the company's emissions intensity relative to production volume.

9.5. Module Design Principles

The Carbon Footprint module should be:

- Structured,
- Easy to use,
- Calculation-oriented,
- Traceable,
- Audit-friendly,
- Scalable for additional categories and methodologies later.

Because this area can become complex, the product should present it in a simple and intuitive way, with clear forms, logical grouping of inputs, and visible calculation outputs.

9.6. Core Functional Requirements

Across all subsections, the module should support:

- Creation of emissions records,
- Editing of emissions records,
- Structured input forms,
- Attachment of supporting documentation,
- Storage of factor source used,

- Storage of calculation result,
- Filtering by period,
- Filtering by location,
- Filtering by source/category,
- Detail view for each record,
- Activity history and traceability.

9.7. Emission Factor Logic

The system should support emissions calculations using:

- Recognized external emission factor sources relevant to the activity type and geography,
- Default public factors where appropriate,
- Optional internal or custom user-defined factors,
- Factor source tracking for transparency and auditability.

Each emissions record should store:

- Emission factor value,
- Factor unit,
- Factor source,
- Factor geography or region,
- Factor year or version,
- Whether the factor is standard or custom.

9.8. Documentation Linkage

Each carbon footprint entry should allow supporting documentation to be attached directly to the record.

These files should also appear in the central Documentation module.

Examples include:

- Electricity bills,
- Fuel invoices,
- Supplier data,
- Transport records,
- Internal spreadsheets,

- Calculation evidence,
- Audit support files.

9.9. Filtering and Overview

The Carbon Footprint module should allow users to filter and review records by:

- Month,
- Quarter,
- Year,
- Plant/location,
- Scope,
- Category,
- Emission source type,
- Factor source.

This is important for both operational review and reporting preparation.

9.10. Relationship with Dashboard and Analysis

The Carbon Footprint module is mainly for structured emissions data entry and record management.

High-level emissions KPIs may appear in the Dashboard, but detailed charts, comparisons, and advanced visual exploration should be handled in the **Analysis** module.

9.11. Product Intent

The Carbon Footprint module should become the company's central operational area for emissions accounting, with a structure that is reliable for daily use and strong enough to support future reporting, analysis, and compliance needs.

10. SCOPE 1

10.1. Purpose

The Scope 1 section should allow the company to register and calculate direct greenhouse gas emissions from sources that are owned or controlled by the company.

This includes operational sources such as fuel combustion, company vehicles, generators, boilers, process emissions, and fugitive emissions where relevant.

10.2. Objective

The objective of this section is to provide a structured way to record direct emissions activity data, calculate the corresponding emissions, and keep supporting evidence linked to each record.

10.3. Scope 1 Record Fields

Each Scope 1 record should include the following fields:

10.3.1 General Information

- Entry Title
- Emission Source Type
- Fuel Type
- Plant / Location
- Month
- Year
- Notes

10.3.2 Activity Data

- Quantity
- Unit
- Source Reference or Consumption Reference, if applicable

10.3.3 Emission Factor Data

- Emission Factor Value
- Emission Factor Unit
- Emission Factor Source
- Emission Factor Region
- Emission Factor Year or Version
- Factor Type (standard or custom)

10.3.4 Calculation Output

- Calculated Emissions
- Output Unit

10.3.5 Supporting Information

- Supporting Documentation
- Created By
- Updated By
- Created At
- Updated At
- Record Status

10.4. Recommended Emission Source Types

The Scope 1 section should support source types such as:

- Stationary combustion
- Mobile combustion
- Company vehicles
- Boilers
- Generators
- Natural gas use
- Diesel use
- LPG use
- Gasoline use
- Process emissions
- Fugitive emissions

The exact list can be expanded later if needed.

10.5. Required Fields for MVP

For the MVP, the following fields should be required:

- Entry Title
- Emission Source Type
- Fuel Type
- Quantity

- Unit
- Month
- Year
- Plant / Location, when applicable
- Emission Factor Source or selected factor
- Calculated Emissions result
- Record Status

10.6. Scope 1 Entry Logic

The user should be able to create a new Scope 1 entry by selecting the source type, entering the activity data, selecting or applying the correct emission factor, and storing the calculated result.
The form should be designed to make the calculation logic clear and easy to follow.

10.7. Scope 1 Overview Page

The Scope 1 list page should provide an organized view of all direct emissions records.

It should include:

- Search bar,
- Filters,
- Create new entry button,
- Result count,
- Table or list view.

10.8. Recommended Filters

The Scope 1 page should support filtering by:

- Month,
- Year,
- Plant/location,
- Source type,
- Fuel type,
- Factor source,
- Record status.

10.9. Recommended List Columns

The list view should display the most relevant information, such as:

- Entry Title
- Emission Source Type
- Fuel Type
- Quantity
- Unit
- Month
- Year
- Plant / Location
- Calculated Emissions
- Factor Source
- Status
- Last Updated

10.10. Scope 1 Detail View

Each Scope 1 record should have a detailed page where the user can view:

- All record fields,
- Supporting documents,
- Calculation details,
- Factor source used,
- Activity history,
- Related tasks,
- Created by,
- Updated by,
- Timestamps.

10.11. Documentation Linkage

Users should be able to attach documents directly to each Scope 1 entry.

Examples may include:

- Fuel invoices,
- Internal logs,
- Meter readings,

- Operational records,
- Transport records,
- Maintenance records,
- Audit evidence.

These documents should also appear automatically in the central Documentation module.

10.12. Traceability Requirement

Because emissions calculations must be auditable, each Scope 1 record should preserve the calculation context used at the time of entry, including the factor source and factor version where possible.

10.13. Product Intent

The Scope 1 section should give the company a reliable and traceable system for managing direct emissions data in a way that is operationally practical and ready for future reporting needs.

11. SCOPE 2

11.1. Purpose

The Scope 2 section should allow the company to register and calculate indirect greenhouse gas emissions resulting from purchased electricity, steam, heat, or cooling consumed by the company.

Its purpose is to give the user a structured and transparent way to monitor energy-related indirect emissions and clearly distinguish between the two main accounting approaches used in Scope 2 reporting.

11.2. Objective

The objective of this section is to allow the company to record energy consumption data, apply the correct emission factors, and calculate both:

- Location-based emissions,
- Market-based emissions.

This section should also ensure that supporting documentation is linked to each record and that the calculation logic remains auditable.

11.3. Scope 2 Record Fields

Each Scope 2 record should include the following fields:

11.3.1 General Information

- Entry Title
- Energy Type
- Plant / Location
- Month
- Year
- Notes

11.3.2 Activity Data

- Quantity
- Unit
- % Renewable
- Energy Provider
- Grid Region

11.3.3 Emission Factor Data

- Emission Factor Value
- Emission Factor Unit
- Emission Factor Source
- Emission Factor Region
- Emission Factor Year or Version
- Factor Type (standard or custom)

11.3.4 Calculation Output

- Location-Based Emissions Value
- Market-Based Emissions Value
- Output Unit

11.3.5 Supporting Information

- Supporting Documentation
- Created By
- Updated By
- Created At
- Updated At
- Record Status

11.4. Required Calculation Logic

The Scope 2 section must calculate and display both:

- Location-based emissions
- Market-based emissions

Both values should be clearly shown to the user so that the difference between the two accounting approaches is visible and understandable.

11.5. Emission Factor Logic

The system should allow the use of:

- Recognized public emission factor sources,
- Region-based electricity factors,
- Supplier-specific factors where available,
- Optional internal or custom emission factors defined by the user.

The selected factor source should always be stored with the record for transparency and traceability.

11.6. Required Fields for MVP

For the MVP, the following fields should be required:

- Entry Title
- Energy Type
- Quantity
- Unit
- Month

- Year
- Plant / Location, when applicable
- Energy Provider, if available
- Grid Region
- Emission Factor Source or selected factor
- Location-Based Emissions Value
- Market-Based Emissions Value
- Record Status

11.7. Scope 2 Entry Logic

The user should be able to create a new Scope 2 entry by entering the energy consumption data, selecting the applicable factors, and storing both the location-based and market-based emissions results.

The form should be structured in a way that makes the distinction between input data, factor logic, and final emissions output very clear.

11.8. Scope 2 Overview Page

The Scope 2 list page should provide an organized overview of all indirect energy emissions records.

It should include:

- Search bar,
- Filters,
- Create new entry button,
- Result count,
- Table or list view.

11.9. Recommended Filters

The Scope 2 page should support filtering by:

- Month,
- Year,
- Plant/location,
- Energy type,
- Energy provider,

- Grid region,
- Factor source,
- Record status.

11.10. Recommended List of Columns

The list view should display the most relevant information, such as:

- Entry Title
- Energy Type
- Quantity
- Unit
- Month
- Year
- Plant / Location
- Energy Provider
- Location-Based Emissions
- Market-Based Emissions
- Factor Source
- Status
- Last Updated

11.11. Scope 2 Detail Page

Each Scope 2 record should have a detail page where the user can view:

- All record fields,
- Supporting documents,
- Calculation details,
- Factor source used,
- Location-based result,
- Market-based result,
- Activity history,
- Related tasks,
- Created by,
- Updated by,
- Timestamps.

11.12. Documentation Linkage

Users should be able to attach documents directly to each Scope 2 entry.

Examples may include:

- Electricity bills,
- Utility invoices,
- Supplier statements,
- Renewable energy documents,
- Internal spreadsheets,
- Meter readings,
- Audit evidence.

These documents should also appear automatically in the central Documentation module.

11.13. Traceability Requirement

Because Scope 2 calculations may depend on different accounting approaches and data sources, each record should preserve the calculation context used at the time of entry, including the factor source, factor version, and whether the result is location-based or market-based.

11.14. Product Intent

The Scope 2 section should provide a clear, traceable, and auditable structure for managing indirect energy emissions, while helping the user understand both location-based and market-based results in a practical and transparent way.

12. SCOPE 3

12.1. Purpose

The Scope 3 section should allow the company to register and calculate indirect greenhouse gas emissions that occur across the value chain and are not included in Scope 1 or Scope 2.

Its purpose is to help the company capture the most relevant indirect emissions sources in a structured, practical, and scalable way, while maintaining traceability and audit readiness.

12.2. Objective

The objective of this section is to provide a usable structure for recording Scope 3 activity data, applying the appropriate calculation method, linking supporting documentation, and generating emissions results for the categories most relevant to the company.

For the MVP, the Scope 3 section should focus on the categories most relevant to a textile manufacturing business.

12.3. Scope 3 Categories Included in the MVP

For the MVP, the recommended Scope 3 categories are:

- Purchased Goods and Services
- Fuel- and Energy-Related Activities not included in Scope 1 or Scope 2
- Upstream Transportation and Distribution
- Waste Generated in Operations
- Business Travel
- Employee Commuting
- Downstream Transportation and Distribution

The platform architecture should still be designed so that additional Scope 3 categories can be added later.

12.4. Scope 3 Structure

Because Scope 3 categories are more diverse than Scope 1 and Scope 2, the section should use a flexible structure with:

- Shared core fields across all Scope 3 entries,
- Category-specific fields depending on the selected category.

This approach keeps the user experience structured while allowing different types of activity data to be recorded.

12.5. Scope 3 Core Record Fields

Each Scope 3 record should include the following common fields:

12.5.1 General Information

- Entry Title
- Scope 3 Category
- Activity Type
- Plant / Location or Business Unit
- Month
- Year
- Notes

12.5.2 Activity Data

- Quantity
- Unit
- Supplier or Counterparty, where applicable
- Activity Reference, where applicable

12.5.3 Emission Factor Data

- Emission Factor Value
- Emission Factor Unit
- Emission Factor Source
- Emission Factor Region
- Emission Factor Year or Version
- Factor Type (standard or custom)

12.5.4. Calculation Output

- Calculated Emissions
- Output Unit
- Calculation Method

12.5.5. Supporting Information

- Supporting Documentation
- Created By

- Updated By
- Created At
- Updated At
- Record Status

12.6. Recommended Calculation Methods

The Scope 3 section should support different calculation approaches depending on data availability.

For the MVP, the platform should support at least:

- activity-based calculation,
- spend-based calculation.

This is important because many companies will not yet have complete physical activity data for all Scope 3 categories.

12.7. Category-Specific Field Logic

In addition to the core fields, each Scope 3 category should allow category-specific inputs.

12.7.1. Purchased Goods and Services

- Supplier
- Material or Service Type
- Quantity or Spend Value
- Currency, if spend-based
- Product Category

12.7.2. Fuel and Energy-Related Activities not Included in Scope 1 or 2

Recommended additional fields:

- Related Energy Type
- Quantity
- Unit
- Upstream Factor Reference

12.7.3. Upstream Transportation and Distribution

Recommended additional fields:

- Transport Mode
- Weight
- Distance
- Origin
- Destination
- Carrier

12.7.4. Waste Generated in Operations

Recommended additional fields:

- Related Waste Flow
- Waste Treatment Path
- Quantity
- Unit
- Waste Operator, if applicable

12.7.5. Business Travel

Recommended additional fields:

- Travel Type
- Transport Mode
- Distance
- Traveler or Department

12.7.6. Employee Commuting

Recommended additional fields:

- Number of Employees
- Transport Mode
- Distance Estimate
- Frequency

12.7.7. Downstream Transport and Distribution

Recommended additional fields:

- Transport Mode
- Weight
- Distance
- Destination
- Carrier

12.8. Required Fields for MVP

For the MVP, the following fields should be required for Scope 3 entries:

- Entry Title
- Scope 3 Category
- Activity Type
- Quantity or Spend Value
- Unit or Currency, depending on method
- Month
- Year
- Emission Factor Source or selected factor
- Calculated Emissions result
- Record Status

Additional required fields may vary depending on the selected category.

12.9. Scope 3 Entry Logic

The user should be able to create a Scope 3 entry by selecting a category first.

Once the category is selected, the form should dynamically show the relevant fields for that category while keeping the shared structure consistent.

The form should clearly separate:

- Activity input,
- Factor input,
- Calculation result,
- Supporting documentation.

12.10. Scope 3 Overview Page

The Scope 3 list page should provide an organized view of all indirect value-chain emissions records.

It should include:

- Search bar,
- Filters,
- Create new entry button,
- Result count,
- Table or list view.

12.11. Recommended Filters

The Scope 3 page should support filtering by:

- Month,
- Year,
- Plant/location,
- Scope 3 category,
- Supplier,
- Activity type,
- Factor source,
- Calculation method,
- Record status.

12.12. Recommended List Columns

The list view should display the most relevant information, such as:

- Entry Title
- Scope 3 Category
- Activity Type
- Supplier or Counterparty
- Quantity or Spend Value
- Month
- Year
- Plant / Location
- Calculated Emissions

- Calculation Method
- Factor Source
- Status
- Last Updated

12.13. Scope 3 Detail Page

Each Scope 3 record should have a detail page where the user can view:

- All record fields,
- Supporting documents,
- Calculation details,
- Factor source used,
- Activity history,
- Related tasks,
- Created by,
- Updated by,
- Timestamps.

12.14. Documentation Linkage

Users should be able to attach documents directly to each Scope 3 entry.

Examples may include:

- Supplier invoices,
- Transport records,
- Purchase documents,
- Travel records,
- Internal spreadsheets,
- Waste records,
- Audit evidence.

These documents should also appear automatically in the central Documentation module.

12.15. Traceability Requirement

Because Scope 3 calculations may use different methods and data quality levels, each record should preserve the full calculation context used at the time of entry, including method, factor source, factor version, and key assumptions where relevant.

12.16. Product Intent

The Scope 3 section should provide a practical and scalable way for the company to start managing indirect value-chain emissions without making the MVP unnecessarily complex, while still being robust enough for future expansion.

13. PRODUCTION EMISSION FACTOR

13.1. Purpose

The Production Emission Factor section should allow the company to calculate its emissions intensity relative to production output.

Its purpose is to help users understand how much greenhouse gas emission is associated with the company's production volume over a selected period, making it easier to monitor operational efficiency and reduction progress.

13.2. Objective

The objective of this section is to combine emissions data from the Carbon Footprint module with production data in order to calculate a clear production-related emissions intensity metric.

This section should help the user monitor performance over time and support internal analysis and reduction planning.

13.3. Core Calculation Logic

The main calculation should be:

Production Emission Factor = Total Emissions / Production Volume

The system should allow the user to calculate this value based on selected emissions scopes and a selected reporting period.

13.4. Inputs

The Production Emission Factor section should include the following inputs:

13.4.1. Emissions Inputs

- Total Scope 1 Emissions
- Total Scope 2 Emissions
- Total Scope 3 Emissions
- Total Combined Emissions
- Scope Inclusion Selection

13.4.2. Production Inputs

- Production Volume
- Production Unit
- Product Type, if applicable
- Plant / Location, if applicable

13.4.3. Reporting Inputs

- Reporting Period
- Calculation Basis
- Notes or Methodology Notes

13.5. Scope Inclusion Logic

The user should be able to choose which scopes are included in the calculation.

For the MVP, the section should allow inclusion of:

- Scope 1 only,
- Scope 1 + Scope 2,
- Scope 1 + Scope 2 + Scope 3.

This is important because some companies may want to view different intensity metrics depending on reporting maturity and internal use case.

13.6. Calculation Output

The section should display:

- Selected total emissions included in the calculation,
- Selected production volume,
- Calculated production emission factor,
- Output unit.

Examples of output units may include:

- kgCO2e per ton of product
- tCO2e per ton of product
- kgCO2e per unit produced
- kgCO2e per meter of textile produced

The output unit should adapt to the production unit selected by the user.

13.7. Record Fields

Each Production Emission Factor record should include:

13.7.1. General Information

- Record Title
- Reporting Period
- Plant / Location
- Product Type, if applicable
- Notes

13.7.2. Emissions Inputs

- Scope 1 Emissions Value
- Scope 2 Emissions Value
- Scope 3 Emissions Value
- Total Emissions Included
- Scope Selection

13.7.3. Production Inputs

- Production Volume
- Production Unit

13.7.4. Output

- Calculated Production Emission Factor
- Output Unit

13.7.5. Supporting Information

- Supporting Documentation
- Created By
- Updated By
- Created At
- Updated At
- Record Status

13.8. Entry Logic

The user should be able to create a Production Emission Factor record by selecting the reporting period, confirming which emissions scopes are included, entering the relevant production volume, and allowing the system to calculate the resulting emission intensity automatically.

Where possible, emissions values should be pulled automatically from the Scope 1, Scope 2, and Scope 3 records for the selected period.

The production volume should be entered manually unless a future integration exists.

13.9. Overview

The Production Emission Factor page should provide an organized overview of all emission factor records.

It should include:

- Search bar,
- Filters,
- Create new record button,
- Result count,
- Table or list view.

13.10. Recommended Filters

The page should support filtering by:

- Reporting period,
- Plant/location,
- Product type,
- Scope inclusion logic,
- Record status.

13.11. Recommended List Columns

The list view should display the most relevant information, such as:

- Record Title
- Reporting Period
- Plant / Location
- Product Type
- Total Emissions Included
- Production Volume
- Production Unit
- Calculated Production Emission Factor
- Output Unit
- Status
- Last Updated

13.12. Detail Page

Each Production Emission Factor record should have a detail page where the user can view:

- All record fields,
- Included emissions values,
- Selected scope logic,
- Production data,
- Final calculated result,
- Supporting documents,
- Activity history,
- Related tasks,
- Created by,
- Updated by,
- Timestamps.

13.13. Documentation Linkage

Users should be able to attach documents directly to each Production Emission Factor record.

Examples may include:

- Production reports,
- Internal spreadsheets,
- Methodology notes,
- Calculation support files,
- Audit evidence.

These documents should also appear automatically in the central Documentation module.

13.14. Trend and Comparison Logic

The section should support viewing the Production Emission Factor over time so the user can monitor whether emissions intensity is improving or worsening.

Detailed graphs and comparative analysis should be handled in the Analysis module, but the Production Emission Factor section should still display the latest calculated value and historical records.

13.15 Product Intent

The Production Emission Factor section should give the company a practical way to connect emissions performance with production output, making the platform more useful for operational decision-making and improvement tracking.

14. ANALYSIS MODULE

14.1. Purpose

The Analysis module should be the advanced visual analytics area of the platform. Its purpose is to allow users to explore sustainability and carbon footprint data in more depth through graphs, comparisons, filters, and visual breakdowns, without overloading the main Dashboard.

14.2. Objective

The objective of this module is to give users a dedicated space to analyze environmental performance, identify trends, compare periods, detect high-impact areas, and better understand the operational drivers behind waste generation and emissions.

14.3. Role in the Platform

The Analysis module should be separate from the Dashboard.

The Dashboard should remain focused on:

- High-level KPIs,
- Recent activity,
- Latest team actions,
- Operational overview.

The Analysis module should be the place for:

- Detailed charts,
- Emissions breakdowns,
- Waste trends,
- Deeper performance exploration,
- Customizable graph views.

14.4. Core Functional Logic

The Analysis module should allow users to:

- View multiple types of graphs,
- Choose which graphs they want to see,
- Filter the data being displayed,
- Compare different time periods,
- Analyze performance by scope, category, plant, or source,
- Switch between different visual views depending on the question they want to answer.

The page should feel configurable and dynamic, rather than fixed and limited to one static set of charts.

14.5. Filtering Capabilities

The Analysis module should support filtering by:

- Date range,
- Month,
- Quarter,
- Year,
- Plant / location,
- Emissions scope,
- Scope 3 category,
- Waste category,
- Waste flow,
- Energy type,
- Fuel type,
- Team member or department, where relevant.

14.6. Graph Selection Logic

Users should be able to select which graphs they want to display on the page.

For the MVP, this can be implemented through a graph selection area where the user adds or removes visual widgets from the screen.

The goal is to give flexibility without making the interface confusing.

14.7. Recommended Graph Types

The Analysis module should support graph types such as:

- Line charts,
- Bar charts,
- Stacked bar charts,
- Area charts,
- Pie charts,
- Donut charts,
- Comparison charts,
- Trend charts,
- Breakdown charts,
- Top contributor charts.

More advanced visualization types can be added later if needed.

14.8. Carbon Footprint Analysis Graphs

The Analysis module should support carbon-related visualizations such as:

- Total emissions over time,
- Emissions by scope,
- Emissions by plant / location,
- Emissions by month, quarter, or year,
- Scope 1 breakdown by fuel type or source,
- Scope 2 comparison between location-based and market-based results,
- Scope 3 breakdown by category,
- Top emissions sources,
- Emissions trend over time,
- Production emission factor trend.

14.9. Waste Analysis Graphs

The Analysis module should also support waste-related visualizations such as:

- Waste volume by category,
- Waste volume by plant / location,
- Hazardous vs non-hazardous waste split,
- Recovery vs disposal split,

- Waste generation trend over time,
- Priority waste flows,
- Top waste streams by volume.

14.10. Data Quality and Completeness Analysis

The module should include analytical views related to data quality and operational completeness, such as:

- Records missing documentation,
- Incomplete emissions entries,
- Waste flows missing required fields,
- Missing classification data,
- Open tasks by module,
- Completion rate by user or team.

14.11. Page Structure Recommendation

The Analysis page should contain:

14.11.1. Top Section

- Page title,
- Filter bar,
- Date range selector,
- Plant/location selector,
- Scope selector,
- Category selector,
- Graph selection controls.

14.11.2. Main Section

- Selected graph widgets,
- Visual cards or chart blocks,
- Configurable graph layout.

14.11.3. Bottom Section

- Optional supporting data table,
- Export actions,

summary notes or insight area in future versions.

14.12. Export Logic

For the MVP, the module should support at least basic export functionality for the selected analysis views or underlying filtered data.

Recommended export options include:

- CSV export,
- Excel export,
- Image or PDF export in future versions.

14.13 UX Recommendation

This module should feel highly visual, flexible, and clean.

Even though it supports many graph types, the interface should remain elegant and intuitive, with strong spacing, clear filters, and easy interaction.

It should feel like a premium analytics workspace, not a cluttered reporting screen.

14.14 Product Intent

The Analysis module should be the place where users go to deeply understand sustainability performance, while the Dashboard remains the simple overview and control center of the platform.

15. DOCUMENTATION MODULE

15.1. Purpose

The Documentation module should be the central repository for all files uploaded to the platform.

Its purpose is to allow users to store, organize, search, filter, preview, and download all supporting documentation related to waste flows, carbon footprint records, production emission factor records, regulations, and other sustainability-related activities.

15.2. Objective

The objective of this module is to centralize all relevant supporting files in one place while maintaining a clear connection between documents and the records they support.

This module should improve traceability, reduce time spent searching for files, and help the company prepare more efficiently for audits and internal reviews.

15.3. Core Functional Logic

The Documentation module should allow users to:

- Upload documents directly,
- View all uploaded documents,
- Search for documents,
- Filter documents,
- Preview supported file types,
- Download documents,
- Link documents to platform records,
- View document metadata,
- Manage document organization in a structured way.

15.4. Supported File Types

For the MVP, the module should support at least the following file types:

- PDF
- Excel files
- CSV files
- image files such as PNG, JPG, and JPEG
- Word files
- text files

Additional file types may be supported later if needed.

15.5. Document Record Structure

Each document uploaded to the platform should include the following metadata fields:

15.5.1. Core Information

- Document Title
- Document Type
- File Name
- File Type
- File Size
- Upload Date
- Uploaded By

15.5.2. Relationship and Context

- Linked Module
- Related Record Type
- Related Record ID
- Reporting Period
- Plant / Location
- Department

15.5.3. Additional Metadata

- Notes or Description
- Tags
- Version Number, if applicable
- Record Status

15.6. Recommended Document Types

The platform should support document classification through a document type field.

Recommended document types include:

- Invoice
- Waste certificate

- Collection receipt
- Fuel bill
- Electricity bill
- Supplier document
- Internal report
- Audit evidence
- Environmental license
- Contract
- Emissions evidence
- Production report
- Regulatory file
- Other

15.7. Documentation Overview Page

The main Documentation page should provide a structured overview of all uploaded files across the platform.

It should include:

- Search bar,
- Filters,
- Upload new document button,
- Result count,
- List or table view,
- Preview action,
- Download action,
- Open related record action.

15.8. Recommended Filters

The Documentation module should support filtering by:

- Document type,
- Linked module,
- Reporting period,
- Upload date,
- Plant/location,
- Department,
- Uploader,

- Tags,
- File type,
- Record status.

15.9. Recommended List Columns

The list view should display the most relevant information, such as:

- Document Title
- Document Type
- Linked Module
- Related Record
- Plant / Location
- Reporting Period
- Uploaded By
- Upload Date
- File Type
- File Size
- Status

15.10. Document Detail View

Each document should have a detail view or side panel where the user can see:

- Full metadata,
- File preview where supported,
- Linked record information,
- Notes or description,
- Tags,
- Version information if used,
- Upload history,
- Download action.

15.11. Document Upload Logic

Users should be able to upload documents in two main ways:

- Directly from the Documentation module,
- Directly from other records, such as Waste Flows, Scope 1, Scope 2, Scope 3, and Production Emission Factor entries.

When a document is uploaded from another module, it should automatically appear in the Documentation module with the correct link to the related record.

15.12. Preview and Download Logic

For supported file types, the user should be able to preview the document inside the platform.

For all supported files, the user should be able to download the original file.

This is important for both operational use and audit preparation.

15.13. Search Logic

The Documentation module should support searching by relevant metadata and file-related information, including:

- Document title,
- Document type,
- Tags,
- Linked record,
- Uploader,
- Notes or description, where indexed.

15.14. Versioning Recommendation

For the MVP, document versioning can be kept simple.

The system should at least allow a version number field or replacement logic if needed, but full advanced version-control workflows do not need to be implemented in the first release.

15.15. Audit and Traceability Value

This module is especially important because it connects source evidence to operational and emissions records.

It should help the company demonstrate:

- Where the supporting evidence is,
- Which record the file belongs to,
- Who uploaded it,
- When it was uploaded,
- Whether the documentation is complete.

15.16. UX Recommendation

The Documentation module should feel highly organized, simple to navigate, and easy to use even when many files exist.

It should avoid clutter and rely on:

- Clean filtering,
- Strong search capability,
- Clear metadata display,
- Intuitive linking to related records.

15.17. Product Intent

The Documentation module should become the company's central evidence library for sustainability operations, helping ensure that all relevant files are accessible, organized, and linked to the correct data records.

16. REGULATIONS MODULE

16.1. Purpose

The Regulations module should provide a centralized area where users can monitor relevant sustainability and environmental regulations that affect the company.

Its purpose is to improve visibility over the regulatory context of the business and support internal compliance awareness.

16.2. Objective

This module should allow users to:

- View relevant EU and country-level regulations,

- Organize regulation-related information,
- Track updates,
- Identify which items are relevant to the company,
- Support internal follow-up.

16.3. MVP Positioning

For the MVP, the Regulations module should function as a curated regulatory information hub, not as a full legal automation engine.

It should help users stay informed, but it should not be positioned as legal advice or as a guarantee of compliance.

16.4. Main Functionalities

The module should allow users to:

- View regulation entries,
- Search and filter regulation items,
- Classify regulations by topic and geography,
- Read simplified summaries,
- See effective dates,
- Mark whether a regulation applies to the company,
- Attach internal notes and supporting files,
- View AI-assisted news or update summaries.

16.5. Module Structure

The module should include two main areas:

16.5.1. Regulation Entries

A structured page where Admin users can manually create, edit, and manage regulation records.

16.5.2. AI-Assisted Updates Area

A separate news or update box that surfaces recent relevant regulatory developments in a summarized way.

This should support awareness, not replace legal review.

16.6. Regulation Record Fields

Each regulation entry should include:

- Regulation Title
- Regulation Type
- Geography
- Topic
- Summary
- Source Reference
- Effective Date
- Status
- Applies to Us flag
- Priority Level
- Internal Notes
- Reviewed By
- Review Date
- Linked Documents
- Created By
- Updated By
- Created At
- Updated At

16.7. Recommended Regulation Types

The module should support types such as:

- EU Regulation
- EU Directive
- National Law
- National Decree
- Guidance
- Reporting Standard
- Regulatory Update
- Internal Compliance Note
- Other

16.8. Recommended Topics

The module should support topics such as:

- Waste Management
- Carbon Footprint
- GHG Reporting
- ESG Reporting
- Energy
- Hazardous Waste
- Environmental Licensing
- Audit and Documentation
- Industrial Compliance
- Other

16.9. Overview Page

The main Regulations page should include:

- Search bar,
- Filters,
- Create new entry button for Admin users,
- Result count,
- List or card view,
- Detail view access.

16.10. Recommended Filters

The module should support filtering by:

- Geography,
- Topic,
- Regulation type,
- Effective date,
- Applies to us flag,
- Priority level,
- Status.

16.11. Detail View

Each regulation entry should have a detail view showing:

- Full summary,
- Source reference,
- Effective date,
- Internal notes,
- Linked documents,
- review information,
- activity history,
- created by,
- updated by,
- timestamps.

16.12. Documentation Linkage

Users should be able to attach files to a regulation entry, such as:

- Official legal documents,
- Internal guidance notes,
- Compliance checklists,
- Interpretation notes,
- Audit support files.

These documents should also appear in the Documentation module.

16.13. Product Intent

The Regulations module should help the company stay organized and informed about the regulatory environment surrounding sustainability operations, while remaining practical and realistic for an MVP.

17. TEAM OVERVIEW

17.1. Purpose

The Team Overview module should provide a centralized view of the people using the platform, their roles, their activity, and their assigned responsibilities.

Its purpose is to improve visibility over team participation, accountability, and operational follow-up within the platform.

17.2. Objective

This module should allow users, especially Admin users, to:

- View all team members,
- Understand each user's role,
- Monitor recent activity,
- Track assigned tasks,
- See open and completed work,
- Improve internal coordination.

17.3. Main Functionalities

The Team Overview module should allow:

- a list of all users in the company account,
- visibility of each user's role,
- visibility of user status,
- visibility of recent activity,
- visibility of open and completed tasks,
- access to user-specific details.

17.4. User Information Display

Each user profile in the Team Overview module should include:

- Full Name
- Email
- Role
- Department, if applicable
- User Status
- Date Invited
- Last Active
- Number of Open Tasks
- Number of Completed Tasks

17.5. Recommended User Status Types

The platform should support user status values such as:

- Active
- Invited
- Inactive
- Suspended

17.6. Team Overview Page

The main Team Overview page should include:

- Search bar,
- Filters,
- User list or card view,
- Summary of team activity,
- Access to user details.

17.7. Recommended Filters

The module should support filtering by:

- Role,
- Department,
- User status,
- Last activity,
- Task status.

17.8. User Detail View

Each user should have a detailed view showing:

- profile information,
- recent actions performed in the platform,
- assigned tasks,
- completed tasks,
- related activity history.

17.9. Activity Tracking

The Team Overview module should display recent user actions such as:

- Records created,
- Records updated,
- Documents uploaded,
- Tasks assigned,
- Tasks completed.

17.10. Product Intent

The Team Overview module should help the company understand who is doing what inside the platform and support better visibility, accountability, and coordination across the team.

18. TASKS AND ACTIVITY TRACKING

18.1. Purpose

The Tasks and Activity Tracking section should allow the platform to support internal follow-up, responsibility assignment, and visibility over operational actions related to sustainability data and workflows.
Its purpose is to make sure that actions are not only recorded, but also assigned, monitored, and completed.

18.2. Objective

This section should allow Admin users to assign tasks in a structured and repeatable way, while allowing all relevant users to track progress, deadlines, and recent activity inside the platform.

18.3. Task Creation Logic

For the MVP, task creation should be simple, clear, and standardized.

An Admin user should be able to create a task using the following inputs:

- Task Title
- Task Description
- Assigned To
- Assigned By

- Related Module
- Related Record
- Priority Level
- Status
- Start Date
- Due Date
- Notes

18.4. Recommended Task Creation Questions

The task creation flow should be designed around a simple logic such as:

- Who is responsible?
- What needs to be done?
- Where is it related on the platform?
- When does it need to be completed?
- How urgent is it?

This makes the task structure intuitive and repeatable.

18.5. Recommended Priority Levels

The platform should support task priority levels such as:

- Low
- Medium
- High
- Critical

18.6. Recommended Task Status Types

The platform should support task status values such as:

- Open
- In Progress
- Blocked
- Completed
- Cancelled
- Overdue

18.7. Task Assignment Logic

Tasks should be assignable to a specific user and, where relevant, linked to a record in the platform, such as:

- a Waste Flow,
- a Scope 1 entry,
- a Scope 2 entry,
- a Scope 3 entry,
- a Production Emission Factor record,
- a Regulation entry,
- a Document.

This ensures that tasks are connected to actual work inside the system.

18.8. Task Overview

The platform should provide a task overview that allows users to see:

- All open tasks,
- Overdue tasks,
- Completed tasks,
- Tasks by assigned user,
- Tasks by module,
- Tasks by priority,
- Tasks due soon.

This can be shown partly in the Dashboard and in more detail through Team Overview.

18.9. Task Detail View

Each task should have a detailed view showing:

- Task title,
- Description,
- Assigned user,
- Assigned by,
- Related module,
- Related record,

- Priority,
- Status,
- Start date,
- Due date,
- Notes,
- Activity history,
- Created at,
- Updated at.

18.10. Activity Tracking Logic

The platform should also track important user activity across modules.

This activity log should help users understand what has happened in the platform and who performed each action.

Tracked activities should include:

- Record creation,
- Record updates,
- Document uploads,
- Task creation,
- Task assignment,
- Task status changes,
- User invitations,
- Regulation updates.

18.11. Activity Log Fields

Each activity log entry should store:

- Activity Type
- User
- Related Module
- Related Record
- Action Description
- Timestamp

18.12. Visibility Logic

Admin users should have full visibility over tasks and activity tracking.

Collaborators should be able to see:

- Tasks assigned to them,
- Relevant task details,
- Their own activity or activity relevant to their work.

Viewer users should have no task editing permissions and only limited visibility where appropriate.

18.13. Notifications Recommendation

For the MVP, notifications can be kept simple.

At minimum, the system should visually show:

- Newly assigned tasks,
- Overdue tasks,
- Recently updated task statuses.

Email or in-app notification workflows can be expanded later.

18.14. Product Intent

The Tasks and Activity Tracking section should make the platform more operational by ensuring that sustainability work is not only recorded, but also assigned, followed up, and made visible across the team.

19. AI Features

19.1. Purpose

The AI Features section should define how artificial intelligence is included in the MVP in a practical and controlled way.

Its purpose is to add useful intelligence to the platform without making the MVP too complex, too expensive, or too dependent on advanced automation from the beginning.

19.2. Objective

The objective of AI in the MVP should be to assist the user, improve usability, and provide helpful insights, rather than trying to automate the full sustainability workflow from day one.

19.3. MVP Recommendation

For the MVP, AI should be included in a lightweight and focused way.

The recommended AI use cases for the first version are:

- In-app assistant,
- AI-assisted regulations/news summaries,
- Basic insight suggestions based on platform data.

19.4. In-App Assistant

The platform should include an AI assistant that helps users interact with the system more easily.

The assistant should be able to:

- Answer questions about how to use the platform,
- Explain module purpose,
- Explain record fields,
- Guide the user on what data is missing,
- Help the user understand calculations at a high level,
- Assist with navigation and platform usage.

This assistant should be positioned as a product-support assistant, not as a compliance expert.

19.5. AI-Assisted Regulations Support

Inside the Regulations module, AI should help surface and summarize relevant updates in a simple and accessible way.

This should include:

- Simplified summaries of relevant updates,
- Support for identifying potentially relevant regulatory developments,
- A news-style update box.

This feature should remain clearly separated from formal legal interpretation.

19.6. Basic Insight Suggestions

The platform may include simple AI or rules-based insight generation to help users identify areas of concern or opportunity.

Examples include:

- Highlighting high-emission sources,
- Pointing out missing documentation,
- Identifying incomplete records,
- Surfacing major contributors to emissions,
- Suggesting areas for reduction review.

For the MVP, this can be relatively simple and does not need to be a highly advanced predictive engine.

19.7. What Should Not be Included in the MVP

To keep the MVP realistic, the following AI capabilities should not be treated as core features in version one:

- Full autonomous compliance engine,
- Legal interpretation engine,
- Complex predictive emissions forecasting,
- Large-scale automatic document extraction,
- Autonomous optimization recommendations,
- Advanced benchmarking against external companies.

These can be explored in later phases.

19.8. Product Positioning of AI

AI should be presented as a useful enhancement layer on top of the platform, not as the only reason the platform works.

The core product value should still come from:

- Structured data,
- Operational visibility,
- Documentation linkage,
- Analysis,
- Collaboration,
- Audit readiness.

19.9. Trust and Transparency

Where AI-generated outputs are shown, the platform should make it clear that the content is supportive and informational.

AI outputs should be reviewable by the user and should not replace human validation in areas such as regulation interpretation or formal reporting decisions.

19.10. Product Intent

The AI layer should make RenAI feel smarter, more modern, and easier to use, while remaining realistic, reliable, and aligned with the needs of an MVP.

20. INTEGRATIONS AND IMPORTS

20.1. Purpose

The Integrations and Imports section should define how data enters the platform from external sources in the MVP.

Its purpose is to reduce manual work where possible and make it easier for companies to start using the platform with their existing files and operational data.

20.2. Objective

The objective of this section is to ensure that the platform supports practical data input methods from the beginning, without making the MVP dependent on heavy technical integrations.

20.3. MVP Recommendation

For the MVP, the platform should prioritize simple and useful import capabilities rather than complex system integrations.

The recommended MVP approach is:

- Manual data entry,
- File uploads,
- CSV import,
- Excel import.

This is the most realistic approach for early customer validation.

20.4. Supported Import Types

The platform should support importing structured data through:

- CSV files,
- Excel files.

This should be available where relevant for modules such as:

- Waste Flows,
- Scope 1,
- Scope 2,
- Scope 3,
- Production Emission Factor,
- Documentation metadata where applicable.

20.5. Import Logic

The import flow should allow the user to:

- Upload a supported file,
- Map imported columns to platform fields,
- Validate the data before final import,
- Identify missing or invalid fields,
- Confirm the import,
- Review imported records after completion.

20.6. Validation Requirements

Before completing an import, the platform should validate:

- Required fields,
- Accepted formats,
- Unit consistency where relevant,
- Duplicate detection where possible,
- Basic data quality issues.

The goal is to prevent incorrect or incomplete data from entering the system.

20.7. File Uploads

In addition to structured imports, the platform should support direct file upload for documentation and supporting evidence across the relevant modules.

This includes uploads from:

- Waste Flows,
- Carbon Footprint records,
- Production Emission Factor,
- Regulations,
- Documentation module.

20.8. Future Integrations

The product architecture should allow future connection to external systems such as:

- ERP platforms,
- Accounting systems,
- Utility providers,
- Internal production systems,
- Document management systems,
- Email ingestion,
- Third-party emissions factor databases.

These do not need to be implemented in the MVP, but the product should be designed so they can be added later.

20.9. Product Intent

The Integrations and Imports section should make the platform practical to adopt from the beginning, while keeping the MVP simple, flexible, and realistic to build.

21. REPORTING AND EXPORTS

21.1. Purpose

The Reporting and Exports section should define how users can extract information and files from the platform for operational use, internal review, and audit preparation.

Its purpose is to ensure that the platform is not only a place to store and analyze data, but also a practical tool for sharing, reviewing, and using information outside the system when needed.

21.2. Objective

The objective of this section is to allow users to export relevant data and download supporting files in a simple and structured way.

21.3. MVP Recommendation

For the MVP, reporting and export functionality should remain practical and focused.

The platform should support:

- Export of structured table data,
- Download of uploaded files,
- Simple filtered data extraction.

Advanced reporting automation can be added later.

21.4. Recommended Export Types

The platform should support export of data from relevant modules such as:

- Waste Flows
- Scope 1
- Scope 2
- Scope 3
- Production Emission Factor
- Documentation metadata
- Tasks, if relevant

21.5. Recommended Export Formats

For the MVP, the platform should support export in the following formats:

- CSV
- Excel

PDF export can be considered later for formatted reports or summaries, but it does not need to be a core MVP requirement.

21.6. File Download Logic

Users should be able to download original uploaded files directly from the platform.

This includes files linked to:

- Waste Flow records,
- Carbon Footprint records,
- Production Emission Factor records,
- Regulations entries,
- Documentation module.

21.7. Export Logic

The platform should allow the user to:

- Export the currently filtered data set,
- Export list views,
- Export structured records,
- Download related source files where available.

This ensures the exports are useful and aligned with what the user is currently viewing.

21.8. Audit and Operational Value

Reporting and export capability is important because users may need to:

- Share data internally,
- Review records outside the system,
- Prepare audit evidence,
- Support reporting workflows,
- Maintain local copies of source documentation.

21.9. Product Intent

The Reporting and Exports section should make the platform practically usable in real business workflows, even if advanced reporting remains limited in the MVP.

22. SECURITY AND COMPLIANCE

22.1. Purpose

The Security and Compliance section should define the minimum legal, technical, and organizational safeguards required for the MVP.

Its purpose is to ensure that RenAI can be used by industrial customers in a credible and secure way, while supporting data protection, access control, and audit readiness from the start. As a multi-tenant B2B SaaS platform handling company data and user data, RenAI should be designed with privacy and security requirements built into the product from the beginning.

22.2. Objective

The objective of this section is to ensure that the platform:

- Protects customer data,
- Limits access according to user role,
- Preserves confidentiality and integrity of records and files,
- Supports traceability,
- Is structured in a way that aligns with GDPR-related obligations for a software provider operating in Europe.

22.3. Multi-Tenant Data Separation

Because RenAI is intended to be multi-tenant, the platform must enforce strict separation between customer accounts.

A user from one company must never be able to access the data, files, users, records, or configuration of another company tenant.

Tenant separation should apply across:

- Application access,
- Database logic,
- File access,
- API responses,
- Search and filtering results,
- Exports and downloads.

22.4. Authentication and Access Control

The platform should include secure authentication and role-based access control.

At minimum, the MVP should support:

- Secure login,
- Password protection with strong password rules,
- Role-based permissions,
- Session protection,
- Restricted access to sensitive actions,
- Controlled access to downloads and documents.

Where possible, the architecture should also allow future addition of features such as single sign-on and multi-factor authentication.

22.5. Core Security Requirements

For the MVP, the platform should include at least the following security controls:

- HTTPS encryption in transit,
- Secure password storage,
- Access control by role and tenant,
- Protected file access,
- Audit logging of key actions,
- Controlled record editing and deletion,
- Backup and recovery process,
- Environment separation between development and production,
- Principle of least privilege for internal access.

GDPR requires appropriate technical and organizational measures and specifically addresses security of processing in Article 32, while processor relationships must be governed by contractual terms under Article 28.

22.6. GDPR and Privacy Requirements

Because the platform will process personal data relating at least to users of the customer company, and may also contain business documents with personal information, GDPR-related controls must be considered from the beginning.

For the MVP, this should include:

- Privacy Policy,
- Terms of Service,
- Data Processing Agreement,
- Clear identification of subprocessors,
- Internal understanding of controller vs processor roles,
- Retention and deletion logic,
- Process for handling data access or deletion requests where applicable.

GDPR also requires records of processing activities in relevant cases under Article 30, and cross-border transfer mechanisms may need to be addressed where data or subprocessors are outside the EEA.

22.7. Document and File Security

Because RenAI includes a Documentation module, file security is a core requirement.

The MVP should ensure that:

- Uploaded files are access-controlled,
- File access follows role and tenant permissions,
- Files are not publicly exposed,
- Document downloads are restricted to authorized users,
- Document actions are traceable.

22.8. Audit and Traceability Requirements

The platform should preserve traceability for important actions.

At minimum, the system should record:

- Who created a record,
- Who updated a record,
- Who uploaded a document,
- When a change was made,
- Task assignments and status changes,
- Important regulation updates,

- Relevant access-related actions where possible.

This is important both for operational accountability and for audit preparation.

22.9. Compliance Positioning

RenAI should not be positioned as automatically making a customer legally compliant.

Instead, the product should be positioned as a platform that:

- Improves data organization,
- Improves traceability,
- Supports audit preparation,
- Supports sustainability operations,
- Helps users monitor relevant regulatory information.

This is a safer and more accurate product position for the MVP.

22.10. Internal Documentation Recommended for RenAI

Before commercial use with customers, RenAI should ideally have the following baseline documentation prepared:

- Privacy Policy,
- Terms of Service,
- Data Processing Agreement,
- Subprocessor list,
- Security overview,
- Internal incident response process,
- Internal backup and recovery process,
- Internal access control policy.

22.11. Product Intent

The Security and Compliance section should ensure that RenAI is built as a credible B2B software product with the minimum protections, controls, and contractual readiness needed to work with industrial customers in Portugal and later in Europe.

23. DATA MODEL AND AUDIT TRAIL

23.1. Purpose

The Data Model and Audit Trail section should define the main entities of the platform and the traceability structure needed to support operational use, reporting, and audit readiness.

Its purpose is to ensure that RenAI is built on a structured data foundation that is scalable, consistent, and capable of preserving accountability for key actions and records.

23.2. Objective

The objective of this section is to define:

- The main entities that must exist in the platform,
- How those entities relate to one another,
- The minimum audit trail requirements for important actions,
- The data structure needed to support the MVP modules.

23.3. Core Data Entities

The MVP should include the following core entities:

- Company
- User
- Role
- Plant / Location
- Waste Flow
- Scope 1 Entry
- Scope 2 Entry
- Scope 3 Entry
- Production Emission Factor Record
- Document
- Regulation Entry
- Task
- Activity Log

These entities should form the main foundation of the platform.

23.4. Company Entity

The Company entity should represent each tenant using the platform.

It should contain company-level data relevant to the SaaS account and serve as the parent structure for users, records, and documents belonging to that tenant.

23.5. User and Role Entities

The User entity should represent each platform user within a company account.

The Role entity should define the permission level of the user, such as:

- Admin
- Collaborator
- Viewer

Users should always belong to a company tenant, and their access should be restricted accordingly.

23.6. Plant / Location Entity

The Plant or Location entity should allow the platform to support site-level organization of sustainability data.

This is important because many industrial businesses operate through one or more facilities, and data may need to be tracked by plant.

Waste, emissions, documentation, and analysis should be able to link to a plant or location where relevant.

23.7. Waste Flow Entity

The Waste Flow entity should store all operational information related to waste streams managed by the company.

It should support classification, quantity tracking, treatment information, notes, hazard indication, and document linkage.

23.8. Carbon Footprint Entities

The Carbon Footprint module should include separate entities for:

- Scope 1 entries,
- Scope 2 entries,
- Scope 3 entries,
- Production Emission Factor records.

These may share some structural logic, but they should remain distinguishable because they contain different business rules and data requirements.

23.9. Document Entity

The Document entity should represent every uploaded file in the platform.

Each document should support metadata, file storage reference, uploader information, and linkage to one or more related records where relevant.

The Document entity should also support module association, document classification, and traceability of upload actions.

23.10. Regulation Entry Entity

The Regulation Entry entity should store manually created or maintained regulation records, including their summary, geography, topic, relevance, and internal follow-up data.

23.11. Task Entity

The Task entity should store operational assignments created inside the platform.

Each task should support assignment, due date, priority, status, linkage to a related module or record, and basic activity history.

23.12. Activity Log Entity

The Activity Log entity should store important user and system actions across the platform.

It should provide traceability and support accountability, audit preparation, and visibility into what has happened inside the system.

23.13. Key Relationships

The data model should support at least the following relationships:

- One Company can have many Users,
- One Company can have many Plants / Locations,
- One Company can have many Waste Flows,
- One Company can have many Scope 1, Scope 2, Scope 3, and Production Emission Factor records,
- One Company can have many Documents,
- One Company can have many Regulation Entries,
- One Company can have many Tasks,
- One Company can have many Activity Log entries.

It should also support relationships such as:

- One User has one Role,
- One User can create many records,
- One Plant / Location can be linked to many records,
- One Document can be linked to one or more module records,
- One Task can be linked to a related module record,
- One Activity Log entry should reference a user and a related module or record where relevant.

23.14. Required Audit Trail Fields

At minimum, important records in the platform should include:

- Created By
- Updated By
- Created At
- Updated At
- Record Status

These fields should be present in operational records such as:

- Waste Flows,
- Scope 1,
- Scope 2,

- Scope 3,
- Production Emission Factor,
- Documents,
- Regulations,
- Tasks.

23.15. Activity Log Requirements

The system should log important actions such as:

- Record creation,
- Record update,
- Document upload,
- Task creation,
- Task assignment,
- Task status change,
- User invitation,
- Regulation update.

Each activity log entry should contain at least:

- Activity Type
- User
- Related Module
- Related Record
- Action Description
- Timestamp

23.16. Data Integrity Principles

The data model should be designed to preserve consistency and reliability across the platform.

Key principles should include:

- Tenant isolation,
- Role-based access to records,
- Consistent linking between documents and records,
- Traceable changes to important data,

- Support for future scalability across multiple customers and additional modules.

23.17. Product Intent

The Data Model and Audit Trail section should ensure that RenAI is built on a structured and scalable data foundation, while preserving the traceability needed for operational trust, collaboration, and audit readiness.

24. NON-FUNCTIONAL REQUIREMENTS

24.1. Purpose

The Non-Functional Requirements section should define the quality standards the platform must meet beyond core features.

Its purpose is to ensure that RenAI is not only functionally correct, but also reliable, secure, usable, scalable, and suitable for professional use by industrial customers.

24.2. Objective

The objective of this section is to define the expected behavior and quality level of the platform in areas such as:

- Performance,
- Usability,
- Reliability,
- Scalability,
- Maintainability,
- Security,
- Responsiveness.

24.3. Performance Requirements

The platform should provide a fast and smooth user experience for normal business use.

For the MVP, this means:

- Pages should load quickly,
- Forms should respond without noticeable delay,
- Filtering and search should be efficient,
- Tables and lists should remain usable with growing data volumes,
- File uploads and downloads should behave reliably.

Performance should remain acceptable even as record volume increases within a tenant.

24.4. Usability Requirements

The platform should be easy to understand and use, especially for non-technical business users.

For the MVP, the interface should be:

- Clear,
- Intuitive,
- Visually organized,
- Easy to navigate,
- Consistent across modules.

The user should not need technical knowledge to complete normal platform actions.

24.5. Design and UX Quality

RenAI should present a modern, premium, and minimalist SaaS experience.

The product should emphasize:

- Strong visual hierarchy,
- Clean layout,
- Good spacing,
- Consistency of components,
- Readable forms and tables,
- Helpful empty states,
- A polished overall feel.

Because design quality is a major priority for the MVP, the platform should avoid clutter, confusion, and visually heavy interfaces.

24.6. Reliability Requirements

The platform should behave consistently and predictably in normal usage.

For the MVP, this means:

- Records should save correctly,
- Calculations should remain stable,
- Uploaded files should remain accessible,
- Filters and list views should return accurate results,
- User actions should be traceable.

The product should minimize unexpected failures in core workflows.

24.7. Scalability Requirements

The platform should be designed to grow beyond the first customer.

For the MVP, the architecture should support:

- Multiple company tenants,
- Multiple users per tenant,
- Multiple plants or locations,
- Increasing record volumes,
- Increasing file volumes,
- Addition of future modules and integrations.

The system does not need to be overengineered, but it should not be built in a way that blocks future expansion.

24.8. Maintainability Requirements

The platform should be structured so that future development and changes are manageable.

The product should favor:

- Modular structure,
- Clean separation of concerns,
- Maintainable code organization,
- Reusable UI components,
- Clear handling of business logic.

This is important because the MVP is expected to evolve after initial validation.

24.9. Security Requirements

The non-functional baseline should include:

- Secure access control,
- Protected file access,
- Safe handling of tenant data,
- Reliable authentication behavior,
- Logging of important actions,
- Protection against common misuse.

Security requirements described in the dedicated Security and Compliance section should be reflected in the implementation quality of the platform.

24.10. Responsiveness Requirements

The MVP should be desktop-first, but the interface should still behave reasonably on smaller screens.

For the MVP:

- Desktop experience is the priority,
- Tablet support should be acceptable where possible,
- Mobile support does not need to be fully optimized for full usage.

A future mobile experience may be developed later, especially for read-only access.

24.11. Availability and Operational Expectations

The platform should be available and usable during normal business operations.

For the MVP, the expectation is not enterprise-grade global infrastructure, but it should still provide:

- Stable hosting,
- Reliable access for customer users,
- Backups,
- Basic recovery capability.

24.12. Product Intent

The Non-Functional Requirements section should ensure that RenAI is built not only as a feature set, but as a credible software product with the quality, usability, and reliability expected by real business customers.

25. TECHNICAL / ARCHITECTURE NOTES

25.1. Purpose

The Technical / Architecture Notes section should define the main technical direction of the platform at a high level.

Its purpose is not to fully prescribe implementation details, but to make clear the architectural expectations needed for the MVP to be scalable, secure, and maintainable.

25.2. Objective

The objective of this section is to ensure that the platform is built with a technical structure that supports:

- Multi-tenant operation,
- Modular growth,
- Secure data handling,
- Future integrations,
- Maintainable development.

25.3. Architecture Direction

RenAI should be built as a multi-tenant web application with a modular architecture.

The system should be structured so that each company tenant has isolated access to its own data, users, documents, and records, while the platform remains centrally managed as one SaaS product.

25.4. Core Architectural Principles

The technical architecture should follow these principles:

- Multi-tenant by design,
- Modular backend structure,
- Clear separation between business domains,
- Scalable data model,
- Secure file storage,
- Role-based access control,
- Auditability of important actions,
- Readiness for future integrations.

25.5. Suggested Business Domains

At a high level, the platform should be organized into the following business domains:

- Authentication and User Management
- Company and Tenant Management
- Waste Flows
- Carbon Footprint
- Documentation
- Regulations
- Team and Tasks
- Activity Logging
- Dashboard and Analysis Aggregation

This structure should help keep the platform maintainable as it grows.

25.6. Database Direction

The platform should use a structured relational data model suitable for business records, permissions, document metadata, and traceability.

The database design should support:

- Tenant isolation,
- User-role relationships,
- Plant/location linkage,
- Record-to-document linkage,
- Task linkage to records,
- Activity logging,
- Emissions calculation records and factor traceability.

25.7. File Storage Direction

Because document management is a core feature, the platform should include secure file storage with controlled access.

The file architecture should support:

- Document upload,
- Storage of metadata,
- Linkage to module records,
- Tenant-based access restriction,
- Reliable download behavior.

25.8. Calculation and Business Logic

The architecture should keep important business logic structured and traceable, especially for:

- Emissions calculations,
- Factor source handling,
- Location-based and market-based Scope 2 calculations,
- Production emission factor calculations,
- Validation of required inputs,
- Task and activity linkage.

Where possible, calculation logic should be centralized and not scattered inconsistently across the product.

25.9. API and Service Structure

The platform should be designed in a way that supports clear API or service boundaries between modules.

This is important for:

- Frontend-backend separation,
- Future integrations,
- Maintainability,
- Scalability of product logic.

25.10. Import and Export Readiness

The architecture should allow support for:

- CSV and Excel imports,
- Structured exports,
- Future third-party integrations,
- Future data ingestion workflows.

Even if these capabilities are simple in the MVP, the architecture should not block their future evolution.

25.11. Environment and Deployment Considerations

The platform should be deployed in a way that supports:

- Separation between development and production environments,
- Secure handling of secrets and configuration,
- Stable hosting,
- Backups and recovery,
- Future scaling as customer usage grows.

25.12. Product Intent

The Technical / Architecture Notes section should ensure that RenAI is built on a professional technical foundation that is realistic for an MVP, but strong enough to support future customers, features, and growth.

26. MVP PRIORITIES AND DELIVERY PHASES

26.1. Purpose

The MVP Priorities and Delivery Phases section should define how the product should be prioritized and structured for implementation.

Its purpose is to help ensure that development focuses first on the highest-value capabilities, while keeping the MVP realistic in scope, strong in quality, and scalable for later evolution.

26.2. Objective

The objective of this section is to:

- Define the main priorities of the MVP,
- Clarify what matters most in delivery,
- Suggest a logical implementation order,
- Help align product ambition with practical execution.

26.3. Main MVP Priorities

The main priorities for the MVP are:

- Strong and beautiful user experience,
- Clean and modern product design,
- Practical usefulness for the customer,
- Reliable core workflows,
- Scalability for future growth.

The product should not feel like a rough internal tool. It should feel credible, polished, and ready to be used by a real industrial customer.

26.4. Core MVP Functional Priorities

From a feature perspective, the highest priorities are:

- User authentication and role structure,
- Multi-tenant company setup,
- Waste Flows module,

- Carbon Footprint module,
- Documentation module,
- Dashboard overview,
- Analysis module,
- Team Overview with task assignment,
- Regulations module in curated form.

These are the capabilities that should define the first usable version of RenAI.

26.5. Suggested Delivery Logic

The MVP should be delivered in a phased way, prioritizing the foundations first and then adding more advanced layers.

A suggested implementation structure is as follows.

26.6. Phase 1 - Core Platform Foundation

This phase should focus on the foundation of the product, including:

- Authentication,
- User roles and permissions,
- Multi-tenant company structure,
- Base navigation,
- Core layout and design system,
- Initial database structure,
- Audit fields and activity foundations.

This phase creates the technical and UX base for everything else.

26.7. Phase 2 - Core Operational Modules

This phase should focus on the most important operational modules:

- Waste Flows
- Documentation
- Scope 1
- Scope 2

These modules create immediate practical value and allow the customer to start entering and organizing real data.

26.8. Phase 3 - Extended Carbon and Analytics

This phase should include:

- Scope 3
- Production Emission Factor
- Analysis module
- Dashboard completion

This phase expands the product into a stronger management and decision-support platform.

26.9. Phase 4 - Team and Regulatory Layer

This phase should include:

- Team Overview
- Task assignment and activity tracking,
- Regulations module,
- AI-assisted regulations updates,
- Basic AI assistant.

This phase strengthens coordination, internal follow-up, and product intelligence.

26.10. MVP Scope Discipline

Although the product has long-term potential, the MVP should remain focused on solving the core operational problem well.

The development process should avoid overcomplicating the first release with:

- Advanced predictive AI,
- Heavy external integrations,
- Overengineered reporting,
- Excessive workflow automation,
- Features that do not directly improve early customer value.

26.11. Post-MVP Expansion Direction

After validation of the MVP, future development may expand into areas such as:

- More advanced AI support,
- Automated document extraction,
- Deeper regulations intelligence,
- External system integrations,
- Read-only mobile experience,
- More advanced reporting,
- Broader industry adaptation.

26.12. Product Intent

The MVP Priorities and Delivery Phases section should ensure that RenAI is built in a way that is ambitious but realistic, polished but focused, and capable of proving value early without losing the longer-term vision.